

EnsDTI: Predicting Drug-Target Interaction With Mixture-of-Experts and Confidence Assessment

Yijingxiu Lu , Soosung Kang , Sun Kim , and Sangseon Lee 

Abstract—Accurately identifying drug–target interactions (DTIs) is a critical step in drug discovery. While structure-based drug design methods demonstrate impressive docking prediction accuracy, their heavy computational demands and resource-intensive nature make them impractical for directly processing vast chemical spaces containing a large number of compounds. This limitation highlights the need for a computational tool that balance speed and accuracy to rank and filter potential drug candidates efficiently. In contrast, existing ligand-based drug design methods, which learn representations from diverse protein and molecule features, often fail to make consistent predictions on unseen data or external databases, limiting their applicability for ranking and filtering potential drug candidates accurately. To address these challenges, we propose EnsDTI, a novel framework that bridges the gap between structure-based and ligand-based drug design approaches. EnsDTI utilizes a mixture-of-experts architecture to enhance DTI predictions using existing deep learning models and incorporates an inductive conformal predictor to assess prediction quality with confidence scores, ensuring reliability. Experimental results on four widely used benchmark datasets show that EnsDTI consistently achieves high performance in both prediction accuracy and confidence estimation. In addition, its candidate rankings correlate well with actual docking affinities, suggesting its practical utility in drug discovery.

Index Terms—Drug target interaction prediction, mixture-of-experts, inductive conformal prediction.

Received 31 December 2024; revised 8 December 2025; accepted 4 January 2026. Date of publication 14 January 2026; date of current version 5 June 2026. This work was supported in part by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education under Grant RS-2023-00246586, in part by the Institute of Information and Communications Technology Planning and Evaluation (IITP) funded by the Korea Government (MSIT) [under Grant RS-2022-00155915, Artificial Intelligence Convergence Innovation Human Resources Development (Inha University)], in part by the Bio & Medical Technology Development Program of the National Research Foundation(NRF), funded by the Ministry of Science and ICT, Republic of Korea under Grant RS-2022-NR067933, in part by the National Research Foundation of Korea(NRF) grant funded by the Korea Government(MSIT) under Grant RS- 2023-00257479, and in part by INHA UNIVERSITY Research Grant. (Corresponding author: Sangseon Lee.)

Yijingxiu Lu is with the Department of Computer Science and Engineering, Incheon National University, Seoul 08826, South Korea (e-mail: solanoon@snu.ac.kr).

Soosung Kang is with the College of Pharmacy, Ewha Womans University, Seoul 03760, South Korea (e-mail: sskang@ewha.ac.kr).

Sun Kim is with the Department of Computer Science and Engineering, Seoul National University, Seoul 08826, South Korea, and also with AIGENDRUG Company Ltd., Seoul 08758, South Korea (e-mail: sunkim.bioinfo@snu.ac.kr).

Sangseon Lee is with the Department of Artificial Intelligence, Inha University, Incheon 22212, South Korea (e-mail: ss.lee@inha.ac.kr).

The source code and datasets are publicly available at <https://github.com/solanoon/EnsDTI>.

This article has supplementary downloadable material available at <https://doi.org/10.1109/TCBBIO.2026.3653990>, provided by the authors.

Digital Object Identifier 10.1109/TCBBIO.2026.3653990

I. INTRODUCTION

ACCURATELY identifying complex drug–target interactions (DTIs) is essential for drug discovery and development [1]. Traditional *in vitro* experimental validation of DTIs is costly and time-consuming, driving the growing reliance on computational methods for more efficient drug discovery [2]. These methods excel at learning latent representations of proteins and small molecules, enabling accurate predictions of their biological properties. Recently, advanced approaches such as diffusion models and large language models have been developed to capture accurate representations of biological properties. Notable examples include ChemBERTa [3] for molecular predictions, ProtBERT [4] and ESMFold [5] for protein structure prediction, AlphaFold 3 [6] for biomolecular interaction predictions, and DiffSBDD [7] for ligand generation conditioned on protein pockets. While highly effective, these structure-based drug design models often suffer from high computational demands, making them impractical for pre-screening large chemical spaces [8], [9], [10]. To overcome this limitation, smaller and more efficient ligand-based deep learning models have been developed to rapidly rank and filter molecular candidates [11], [12]. By leveraging diverse pharmacological, genetic, and sequential features, these models achieve competitive performance across multiple datasets while maintaining lower computational costs [13], [14], [15], [16], [17].

Despite notable advances, ligand-based deep learning models for DTI prediction face two critical challenges: inconsistent performance across datasets and the absence of prediction confidence measures [18]. Our preliminary study on existing models (Fig. 1) reveal key issues: (1) predictions from different DTI expert models often lack consensus, yielding divergent results for identical test samples; (2) the same model’s performance can vary substantially across data sources. These inconsistencies undermine the reliability of computational methods in guiding downstream drug development processes. Furthermore, most models do not quantify prediction confidence, leaving researchers uncertain about the reliability of their results when screening potential drug candidates. Addressing these challenges by improving prediction accuracy and providing reliable confidence measures is essential for prioritizing compounds and optimizing experimental validation efforts.

Recent multi-modal architectures have further advanced sequence- and structure-based DTI prediction. Notable examples include DeepTGIN [19] and MMPD-DTA [20], which integrate heterogeneous information such as molecular graphs,

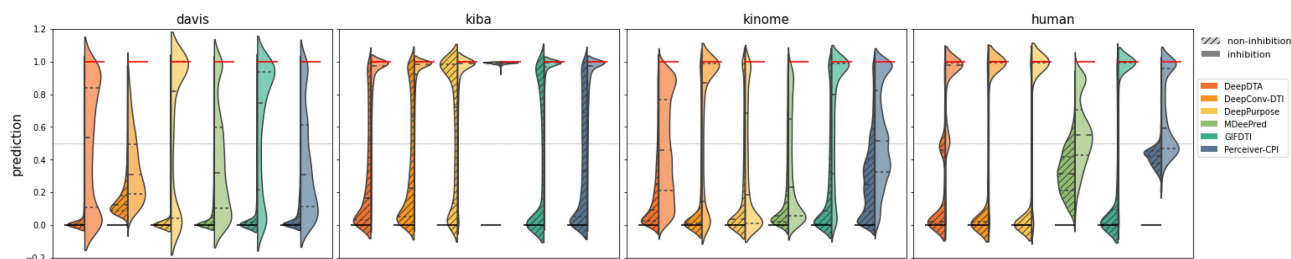


Fig. 1. Distribution of prediction scores across benchmark datasets reveals inconsistency among DTI models. To investigate the variation in model behavior across benchmark datasets, we analyzed the distribution of prediction scores generated by each DTI model using violin plots. In each subplot, the y-axis denotes the predicted probabilities (ranging from 0 to 1) for binary classification, with the left and right halves representing positive and negative samples, respectively. The results indicate substantial differences in prediction patterns both across datasets and within individual models. For example, in the KIBA dataset (with an approximate positive-to-negative ratio of 4:1), models such as DeepPurpose and MDDeePred frequently assign high scores to negative samples, suggesting a tendency toward false positives. Conversely, in the Davis dataset, several models assign low probabilities to positive samples, implying a higher rate of false negatives. These observations reflect the lack of consistent generalization performance among current DTI models across diverse evaluation settings.

protein sequences, and structural descriptors to improve feature fusion and representation capacity. Although they achieve strong performance by leveraging multi-modal information, they still do not sufficiently address robustness under data inconsistency and the lack of calibrated confidence estimates.

To address the limitations of current deep learning-based DTI models, we propose EnsDTI, a mixture-of-experts (MoE) framework designed to enhance both the accuracy and reliability of drug-target interaction (DTI) prediction across heterogeneous datasets. EnsDTI follows a four-stage training pipeline. In the first stage, each expert model is independently trained using its designated input features to perform DTI prediction. Once trained, the expert models are fixed, and their prediction probabilities are used as inputs to a gating network. In the second stage, the gating network learns to combine these outputs through adaptive weighting, aiming to tailor the ensemble prediction to the specific characteristics of each input pair. The third stage involves fine-tuning the ensemble to improve its generalization capacity. Finally, an inductive conformal prediction (ICP) module is applied to assign calibrated confidence scores, enabling more interpretable and robust candidate ranking.

The main contributions of this work are summarized as follows:

- We introduce EnsDTI, a mixture-of-experts framework that integrates multiple DTI models to improve predictive performance and robustness across diverse datasets.
- EnsDTI incorporates an inductive conformal prediction (ICP) module that provides calibrated confidence estimates, supporting more informative and trustworthy ranking of candidate interactions.
- Extensive experiments on four benchmark datasets demonstrate that EnsDTI consistently outperforms individual expert models in terms of both predictive accuracy and score distribution consistency.
- Case studies show that EnsDTI yields more stable candidate rankings for kinase proteins with similar biological functions, suggesting improved generalizability to related targets.

These findings suggest that EnsDTI may help address inconsistencies in DTI prediction and support more efficient candidate

selection in early-stage drug discovery. To support reproducibility and facilitate further research, the source code and datasets are available at <https://github.com/solanon/EnsDTI>.

II. RELATED WORKS

Accurately identifying drug-target interactions (DTI) is crucial for accelerating drug discovery [1]. Deep learning-based DTI models typically rely on protein and drug representations, such as protein amino acid sequences and small molecule SMILES strings. These models employ diverse architectures and feature encoding techniques to predict DTI, focusing on different aspects of the drug-protein interaction [21].

A. Sequence and Structure-Based Models

Models in this category utilize protein amino acid sequences and small molecule SMILES strings or graph structures to encode drug and target information. DeepDTA [22] employs convolutional neural networks (CNNs) to extract features from both protein sequences and drug SMILES strings. By concatenating these representations, it predicts drug-target binding affinities. DeepConv-DTI [23] also utilizes CNNs with varying window sizes to capture local residue patterns from protein sequences, while representing drugs using extended-connectivity fingerprints (ECFP). DeepPurpose [24] introduces a modular framework, combining CNNs for protein encoding with message-passing neural networks (MPNNs) for extracting chemical graph features. These models demonstrate strong predictive performance but may struggle with generalization across diverse datasets. GIFTI [25] enhances feature extraction by integrating CNNs and Transformers, capturing both local residue-level patterns and global sequence-wide features. This hybrid approach enables the modeling of long-range dependencies critical for accurate DTI prediction.

B. Multi-Feature and Multi-Modal Models

To improve DTI prediction accuracy, several models incorporate multiple data modalities and feature types, such as evolutionary, physicochemical, and structural properties.

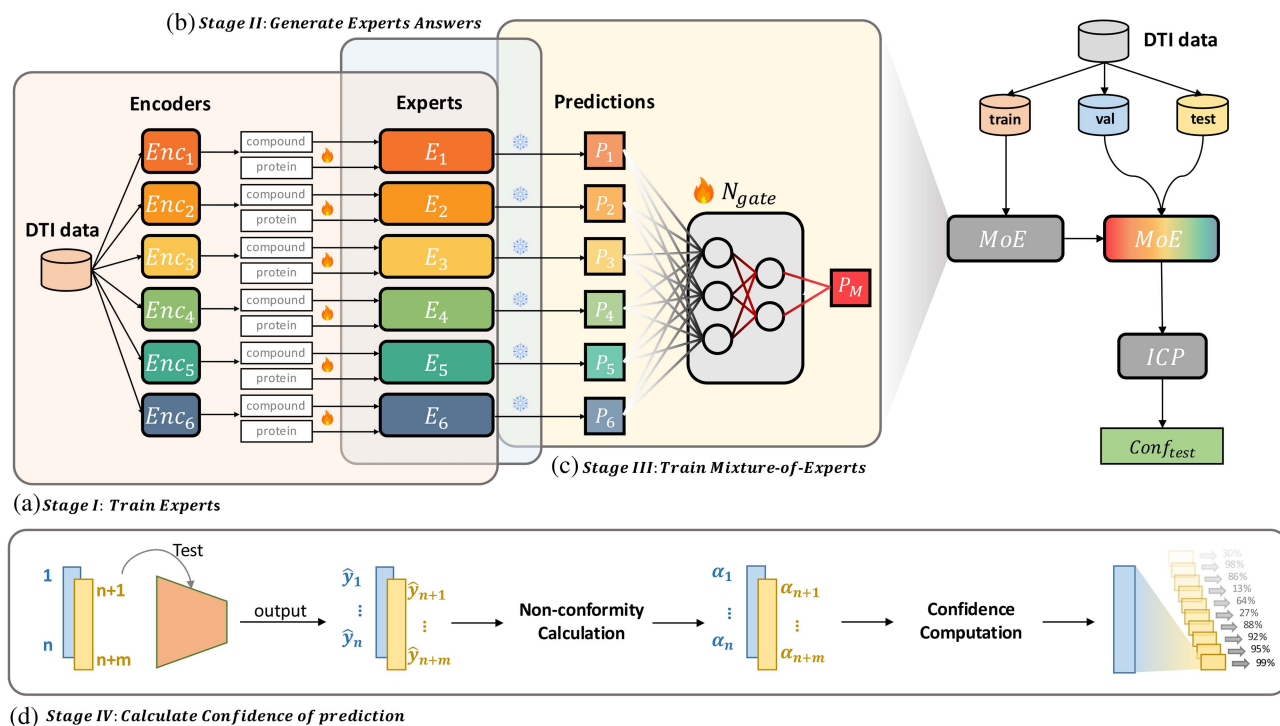


Fig. 2. Overview of EnsDTI: The training and testing process of EnsDTI consists of four stages: (a) Train experts: DTI expert models are re-trained on individual databases to specialize in specific datasets. (b) Generate expert predictions: After training, the parameters of all experts are fixed, and their predictions are generated as input for the mixture-of-experts (MoE) training. (c) Train MoE: A gating network is trained using the experts' predictions to produce the final MoE output. (d) Calibrate predictions: An inductive conformal predictor is applied to the MoE output to calculate confidence values for each sample.

MDeePred [26] combines diverse protein features, including evolutionary information and physicochemical attributes, providing a comprehensive representation for affinity prediction. PerceiverCPI [27] leverages cross-modal attention mechanisms to identify relevant subsequences and substructures within both drug and protein representations, enabling it to model complex interactions. These approaches aim to integrate richer information but often face challenges in balancing model complexity and interpretability.

Recent multi-modal DTI methods aim to expand the representation space by jointly modeling heterogeneous molecular and protein information. DeepTGIN [19] employs a hybrid multi-modal design, using two transformer encoders to extract global and pocket-level protein sequence features and a GIN encoder to capture ligand structural information, enabling joint learning of sequential and graph modalities. MMPD-DTA [20] introduces a multi-modal dual-scale framework that fuses global protein/drug representations with a dedicated pocket-drug interaction graph, allowing explicit modeling of atomic-level binding-site relations for more mechanistically grounded affinity prediction. Although these approaches significantly enhance representational richness, they mainly focus on multi-modal feature integration and do not directly address challenges such as prediction variability, or confidence calibration. In contrast, our work emphasizes ensemble-based mixture-of-experts learning and inductive conformal prediction to improve robustness, reliability across heterogeneous datasets.

C. Challenges in Existing DTI Models

While these models exhibit strong predictive capabilities, our preliminary studies highlight several key challenges. First, their reliance on specific architectural designs and feature encoding methods often results in inconsistencies across datasets. Additionally, the independent nature of their learned latent spaces can lead to divergent predictions for the same test data. Such variability limits their utility in high-stakes drug development applications.

III. METHODS

To address the above limitations, we propose EnsDTI, a Mixture-of-Experts (MoE) architecture that integrates the strengths of diverse expert models. By leveraging inductive conformal prediction, EnsDTI not only provides robust predictions but also quantifies confidence in its results, ensuring higher reliability for downstream applications. The overall pipeline of our MoE model, EnsDTI, is depicted in Fig. 2. The total training and testing process could be separated into four stages: (I) train experts; (II) generate experts answers; (III) train MoE, and (IV) calculate confidence of prediction. illustrating the complete process of boosting and stabilizing current DTI prediction experts with our method. In this section, we will describe in detail: (1) how we train EnsDTI with DTI expert models (stage I to III), and (2) how we use inductive conformal prediction (ICP) to provide confidence values for prediction (stage IV).

A. Problem Formulation

Given a pair of drugs $d \in D$ and a protein $p \in P$, the objective of DTI task is to predict their binding label $y \in Y$, $Y = \{\text{'binding'}, \text{'non-binding'}\}$. In the context of deep learning-based DTI models, a non-linear function $f : D \times P \rightarrow Y$ is constructed to predict binding labels from Y of drug-protein pairs from D and P . Hence, we denote a set of DTI expert models as $F : f_1, f_2, \dots, f_n$. In this work, we propose to mix the experts with a gating network, denoted as g , which operates on the existing functions F , to map from F to Y . This can be written as $g(f_1, f_2, \dots, f_n) \rightarrow Y$.

B. EnsDTI: Mixture of DTI Experts

Mixture of experts (MOE) is a powerful machine learning technique that leverages multiple specialized models to enhance overall prediction accuracy and robustness. By integrating the strengths of various models, MoE effectively captures different aspects of the data, leading to improved performance. This approach also mitigates overfitting and reduces the risk associated with relying on a single model, which may have inherent limitations or biases [28], [29].

In constructing EnsDTI, we utilized six state-of-the-art (SOTA) DTI experts: DeepDTA, DeepConv-DTI, DeepPurpose, MDDeePred, GIFDTI, and PerceiverCPI. As illustrated in Fig. 2(a), each DTI expert was trained with the reported optimal parameters using a 5-fold cross-validation setting. Upon completion of the training process, the parameters of all experts were fixed. We then generated soft label predictions from these models, representing their confidence in the interaction between a given compound and protein pair (Fig. 2(b)).

To further enhance the model, we introduced a multilayer perceptron (MLP)-based gating network g that maps from F to Y (Fig. 2(c)). Formally, this can be represented as follows:

$$\text{MoE}(d, p) = \text{MLP}([f_1(d, p), f_2(d, p), \dots, f_5(d, p)]) \quad (1)$$

Consequently, EnsDTI incorporates the insights from individual DTI experts to enhance the overall predictive performance.

C. ICP-Based Confidence Measurement

However, no model can accurately predict all samples. Providing confidence levels for each prediction based on the MoE model can further assist researchers in identifying small molecules that are likely to bind to target proteins. The ICP framework, which independently estimates the credibility of model predictions, can be readily applied to assess the confidence levels of each sample [30]. For each test sample, ICP predicts the likelihood of assignment to each available class by comparing its prediction with those from the training dataset.

Therefore, we incorporate the ICP framework [30] into EnsDTI to assess the confidence levels of model predictions. Specifically, let X denote the full dataset comprising the drug set D , target set P , and their interactions Y . ICP divides X into three subsets: the training set X_{train} used for model training, the calibration set X_{calib} with known labels, and the test set X_{test} with unknown labels. The confidence value for each test sample

TABLE I
FOUR BENCHMARK DATASETS USED FOR EVALUATING DTI PERFORMANCE. P/N REPRESENTS THE RATIO OF POSITIVE TO NEGATIVE LABELS IN THE ENTIRE DATASET.

Data	Samples	P/N	drugs	targets
davis	28,284	1/11	64	442
kiba	118,254	4/1	2,090	229
kinome	48,547	1/1.6	28,860	346
human	6,212	1.2/1	2,726	2,011

is then measured by comparing the non-conformity score α_i of the test sample to the non-conformity scores of the calibration samples in X_{calib} .

The non-conformity score α_i for each sample x_i from either X_{calib} or X_{test} is calculated as follows:

$$\alpha_i = 0.5 - \frac{P(y_i|x_i) - \max_{y \neq y_i} P(y|x_i)}{2} \quad (2)$$

where $P(y_i|x_i)$ is the predicted probability of the true label y_i given the input x_i . The non-conformity score measures the dissimilarity between each test prediction and the predictions on the calibration set. This calculation is performed for each possible label y_i in Y according to Equation 2, evaluating the suitability of classifying the test sample into each corresponding class.

Finally, to determine the prediction with the highest confidence, we compute the p -value as:

$$p(y|x) = \frac{\sum_{x_i \in X_{\text{calib}}} \mathbb{I}(\alpha_i \geq \alpha)}{|X_{\text{calib}}|} \quad (3)$$

where α represents the non-conformity score of the test sample x . Each test sample is then assigned to the class with the highest p -value, indicating the predicted label with the greatest confidence.

IV. EXPERIMENTS

A. Data Pre-Processing

To fairly evaluate EnsDTI with baseline models, we use four benchmark databases that generated from different experiment settings to make sure each database has a distinct distribution of data.

- *davis* [31] contains data from kinase selectivity assays with their relevant inhibitors measured in K_d .
- *kiba* [32] is constructed by combining data with different types of bioactivity (IC_{50} , K_i , and K_d).
- *kinome* [26] is generated from ChEMBL (v25) by filtering representative compounds.
- *human* [33] contains interactions across different species with the same number of highly credible negative samples of each compound-protein pair.

To adapt the data for our task and ensure that the data can be utilized by each model, we removed compounds with non-covalent bond, and binarized the affinity values according to [32]. Detailed information (e.g. the positive-negative ratio, number of samples) about the data is listed in the Table I.

We evaluated EnsDTI by comparing it with the DTI expert models used in our framework under the previously mentioned

DTI benchmark databases: davis, kiba, kinome, and human. Under each database, we employed the k-fold cross-validation setting for all the expert models and EnsDTI. Specifically, we randomly partition each database into k subsets and conducting experiments k times. In each iteration, one subset is designated as the test set to evaluate the model's performance, while the remaining k-1 subsets are used for model training. By repeatedly reshuffling the data and rotating the test set among the subsets, k-fold cross-validation provides a comprehensive assessment of the model's performance on different subsets of the data.

To ensure the validity of our model, we randomly sampled and split 10% of the training data to create a validation set. This validation set is used to tune hyperparameters and make informed decisions during model training. In this study, we set k to 5, which means the dataset was divided into five equally sized subsets, and the training and evaluation process was repeated five times, each time using a different subset as the test set.

B. Evaluation Metrics

To evaluate the performance of our model, we employed three widely used metrics in drug-target interaction prediction: accuracy (ACC), area under the receiver operating characteristic curve (AUROC), and area under the precision-recall curve (AUPRC). These metrics play a crucial role in assessing the predictive capabilities of computational models in this domain.

The ACC measures the overall correctness of the predictions. It is computed as the ratio of the number of correctly predicted interactions to the total number of interactions in the dataset. Mathematically, ACC can be expressed as:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

The AUROC is a widely used metric for evaluating the model's ability to distinguish true interactions from non-interactions across different decision thresholds. It measures an area under the ROC curve, which consists of the true positive rates (sensitivity) against the false positive rates (1-specificity) at various threshold values. AUROC ranges from 0 to 1, with higher values indicating better predictive performance. The AUROC can be calculated as:

$$AUROC = \int_0^1 TPR(fpr) d(fpr) \quad (5)$$

where TPR represents the true positive rate and fpr represents the false positive rate.

Another important metric for drug-target interaction prediction is the AUPRC. It provides insight into the model's ability to prioritize true interactions over non-interactions, particularly in scenarios where the dataset is imbalanced. The AUPRC is calculated by integrating the precision-recall curve and is given by:

$$AUPRC = \int_0^1 Precision(recall) d(recall) \quad (6)$$

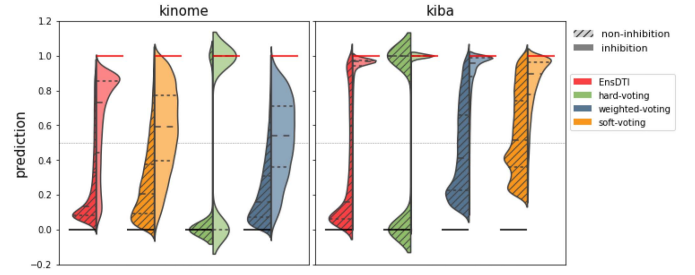


Fig. 3. We compare EnsDTI with traditional voting-based methods to assess its effectiveness of leveraging expert knowledge to refine decision boundaries. Each violin plot represents the distribution of prediction values, with the left and right portions corresponding to positive and negative samples, respectively. Additional details are provided in Section V-A2.

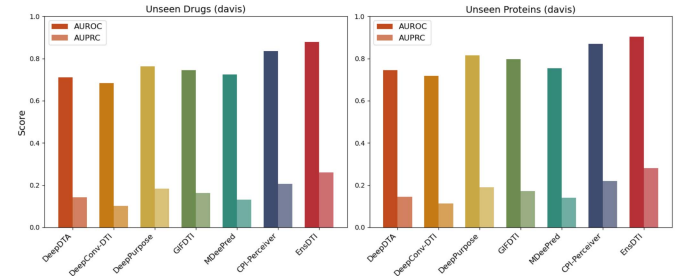


Fig. 4. Performance of EnsDTI and baseline models on unseen data (Davis dataset). Left: unseen drugs (cold drug) setting; Right: unseen proteins (cold protein) setting. Results are presented in terms of AUROC and AUPRC, illustrating EnsDTI's robustness under cold-start scenarios.

V. RESULTS

A. Overall Performance

1) *MoE Architecture Improves the Performance of DTI Prediction*: We conducted a comprehensive comparison between EnsDTI and existing DTI expert models using a 5-fold cross-validation setting across datasets with varying data distributions. The results, presented in Table II, demonstrate that EnsDTI significantly enhances the performance of current DTI models across all evaluated databases.

Notably, even in challenging scenarios where certain experts exhibit unreliable performance—such as in the Davis dataset, where three out of six expert models report an AUPRC value of 0.5 or lower—EnsDTI maintains strong predictive accuracy, achieving an AUPRC of 0.670. This highlights its capacity to effectively harness the strengths of individual experts while mitigating their weaknesses, enabling more reliable and accurate predictions.

In addition, we followed the experimental protocol of [34] to evaluate performance under unseen drugs (cold drug) and unseen proteins (cold protein) settings on davis dataset. As shown in Fig. 4, EnsDTI consistently outperforms baseline models under these cold-start conditions.

Table II underscores the specialization of existing DTI expert models, where each model tends to perform well on specific datasets but struggles on others. By integrating these diverse expertise areas through its MoE framework, EnsDTI balances

TABLE II

PERFORMANCE COMPARISON OF THE PROPOSED MIXTURE-OF-EXPERTS ARCHITECTURE (ENSDTI) AND EXISTING DTI MODELS UNDER CROSS-VALIDATION. THE MODEL WITH THE BEST OVERALL PERFORMANCE IS HIGHLIGHTED IN BOLD, WHILE THE BEST-PERFORMING EXPERT MODEL IS UNDERLINED.

Model	davis			kiba			kinome			human		
	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC
DeepDTA	0.938	0.901	0.603	0.875	<u>0.915</u>	<u>0.955</u>	0.730	0.802	0.716	0.893	0.888	0.921
DeepConv-DTI	0.924	0.828	0.440	<u>0.890</u>	<u>0.909</u>	<u>0.953</u>	<u>0.767</u>	<u>0.821</u>	0.757	0.906	0.905	0.938
DeepPurpose	0.939	0.923	0.636	0.773	0.715	0.910	0.610	0.602	0.550	0.908	0.910	0.943
GIFDTI	0.927	0.912	0.604	0.886	0.914	0.944	0.742	0.802	0.857	0.905	0.970	<u>0.976</u>
MDeePred	0.929	0.869	0.500	0.790	0.599	0.895	0.670	0.696	0.627	0.763	0.773	0.859
CPI-Perceiver	0.928	0.863	0.488	0.877	0.901	0.946	0.714	0.762	0.826	0.825	0.904	0.920
EnsDTI	0.949	0.929	0.670	0.906	0.942	0.958	0.795	0.866	0.790	0.959	0.958	0.977

TABLE III

PERFORMANCE COMPARISON OF THE PROPOSED MIXTURE-OF-EXPERTS ARCHITECTURE (ENSDTI) WITH TRADITIONAL VOTING-BASED METHODS UNDER CROSS-VALIDATION. HERE, “SOFT” DENOTES SOFT VOTING, “HARD” DENOTES HARD VOTING, AND “WEIGHTED” DENOTES WEIGHTED VOTING. THE MODEL ACHIEVING THE BEST OVERALL PERFORMANCE IS HIGHLIGHTED IN BOLD. FURTHER DETAILS CAN BE FOUND IN SECTION V-A2.

Model	davis			kiba			kinome			human		
	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC
Soft	0.948	0.934	0.672	0.874	0.935	0.933	0.777	0.849	0.768	0.947	0.976	0.960
Hard	0.944	0.755	0.639	0.882	0.737	0.938	0.765	0.735	0.756	0.945	0.945	0.965
Weighted	0.948	0.935	0.659	0.897	0.940	0.948	0.779	0.861	0.776	0.955	0.990	0.973
EnsDTI	0.949	0.929	0.670	0.906	0.942	0.958	0.795	0.866	0.790	0.959	0.958	0.977

the knowledge contributions of each expert using its gating network. This design ensures stable and robust predictions, even when applied to datasets with varying characteristics, further validating the generalizability of our approach across diverse data distributions.

2) *EnsDTI Shows Stable Prediction Across Different Distribution of Data When Compared With Voting Methods:* The MoE framework can be understood as a specialized voting mechanism that dynamically learns decision boundaries for aggregating predictions through a trained gating network. To assess whether this automated approach outperforms traditional methods for combining predictions from DTI expert models, we compared it against conventional voting schemes, including **soft-voting**, **hard-voting**, and **weighted-voting**, across all datasets. Specifically:

- The **soft-voting method** calculates the final prediction by averaging the predicted probabilities from the six experts.
- The **hard-voting method** determines the label prediction based on the majority vote of the six experts.
- The **weighted-voting method** computes the final prediction by summing the weighted accuracy of each expert model and its predicted label.

Table III summarizes the predictive performance of our MoE framework compared to three traditional voting methods. The results indicate that while traditional voting approaches generally enhance performance by aggregating predictions from expert models, they consistently underperform compared to EnsDTI. Notably, hard voting demonstrates lower AUROC across different datasets, highlighting its limitations.

Fig. 3 provides a detailed comparison of the prediction probability distributions generated by our MoE framework and the three voting methods. The findings reveal that traditional voting schemes are strongly influenced by the individual predictions of expert models. For instance, as shown in the top panels, under both the Davis and Kinome datasets, all three voting methods excel in predicting positive labels but tend to generate

a significant number of false positives. In contrast, EnsDTI, guided by its learned gating network, achieves higher precision by effectively reducing false positive rates.

Additionally, the bottom-left panel of Fig. 3 shows that EnsDTI tends to produce fewer false negatives when aggregating predictions from DTI expert models. This results in higher recall compared to traditional voting methods, contributing to more comprehensive identification of true interactions.

B. Results on Confusing Data

For certain challenging or deceptive data points, how accurately can the mixture-of-experts mechanism ensure predictions when the majority of expert models provide incorrect answers? To intuitively assess EnsDTI’s performance on these difficult data points, we graded the difficulty of all test data and plotted the performance of each expert model and EnsDTI across different difficulty levels.

we categorized the test data into varying difficulty levels and compared the performance of EnsDTI and individual expert models across these levels. Data difficulty was graded based on the collective error rates of the expert models, with the intuition that data points misclassified by all six experts are more difficult than those where only one expert errs.

Fig. 5 presents the test accuracy (ACC) of EnsDTI and baseline models across subsets of differing difficulty. The horizontal axis of each subplot denotes the number of experts correctly predicting the subset, while the vertical axis indicates the accuracy of each model on that subset. The test data were partitioned into seven levels (x-axis values from 0 to 6), with higher values indicating lower difficulty and greater ease. Level 0 corresponds to the most challenging subset (where all six experts failed), while Level 6 represents the easiest subset (where all six experts succeeded).

From Fig. 5, we observe that, in most datasets, EnsDTI achieves the largest area under the curve and maintains relatively

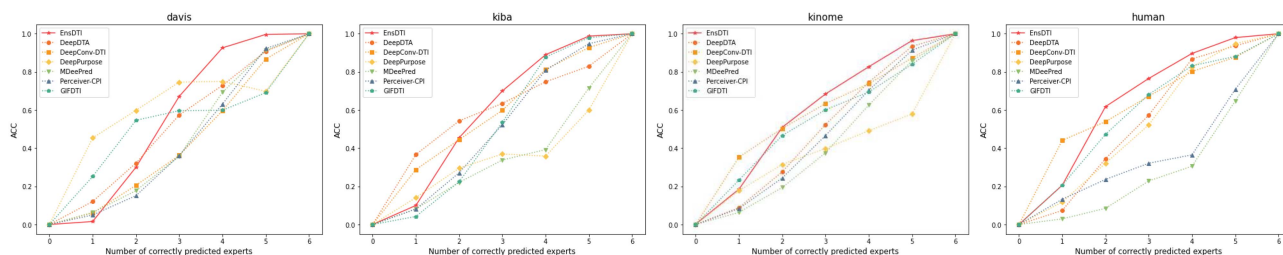


Fig. 5. Degree of prediction consensus of EnsDTI and baseline models on data subsets with varying difficulty. Test data were divided into seven difficulty levels based on the number of correctly predicting experts (0–6). The horizontal axis indicates the number of correct predictions by experts (higher values mean easier subsets), and the vertical axis shows the test accuracy (ACC). EnsDTI consistently outperforms baseline models across most levels, especially in subsets from Levels 2 to 6, demonstrating its robustness and reliability on challenging datasets.

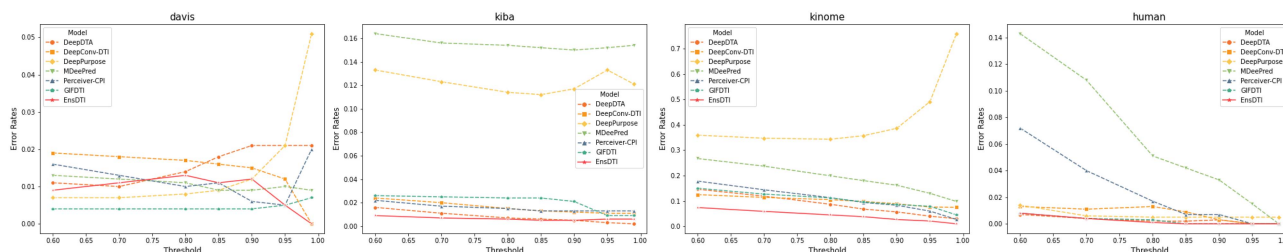


Fig. 6. Error rates of EnsDTI and baseline models across confidence intervals. Confidence levels were computed using the ICP mechanism, and test samples were grouped by confidence thresholds. EnsDTI demonstrates consistently lower error rates than baseline models in most confidence intervals, with notable improvements for confidence levels exceeding 0.9. In three datasets, EnsDTI achieves an error rate below 1% for confidence levels above 0.6, while maintaining competitive performance in the remaining dataset. These results highlight EnsDTI's reliability and the effectiveness of ICP in enhancing confidence-based predictions.

high accuracy across all subsets from levels 2 to 6. Notably, when 4 out of 6 experts correctly predict a sample, EnsDTI reaches over 80% accuracy, exceeding the corresponding values of the individual models. These findings suggest that EnsDTI can benefit from expert agreement and may offer improved robustness on challenging cases.

C. Results on Confidence

As described in Section III-C, we leverage the ICP mechanism to provide confidence evaluations for each prediction generated by EnsDTI. Our goal is to assess whether EnsDTI demonstrates higher confidence in its predictions compared to existing DTI expert models, and to examine whether this enhanced confidence correlates with improved accuracy. To this end, we analyze the error rates of EnsDTI and six DTI expert models across test subsets partitioned by confidence intervals.

Fig. 6 illustrates the error rates of EnsDTI and baseline models across various confidence interval subsets. Confidence intervals are computed using ICP, and test samples with confidence values exceeding 0.6 are grouped accordingly. EnsDTI consistently exhibits lower error rates than the baseline models across most confidence intervals, with particularly significant improvements observed for confidence levels above 0.9.

In three out of the four datasets, EnsDTI achieves an error rate below 1% when its prediction confidence exceeds 60%. In the remaining dataset (kiba), EnsDTI still maintains a lower error rate than other models, staying under 10%. While EnsDTI's error rate in the davis dataset slightly surpasses that of GIFDTI

and PerceiverCPI for some confidence levels, it achieves the lowest error rate for samples with confidence levels exceeding 95%. Overall, as EnsDTI's confidence in its predictions increases, its error rate steadily declines. These findings not only demonstrate the utility of ICP in enabling researchers to effectively use EnsDTI for screening potential drug candidates but also highlight EnsDTI as a more reliable model compared to other approaches across most datasets.

D. Potential on Inferring Bioactivity

Although models trained on binarized labels are not inherently designed to predict precise affinity values, their prediction probabilities can still implicitly indicate the likelihood of how effectively a small molecule targets a specific protein. To evaluate EnsDTI's ability to infer and approximate true binding strength, we conducted the following experiment.

Using the kinome dataset, we calculated the Pearson correlation between each model's prediction probabilities and the corresponding pChEMBL values, as shown in Fig. 7. The pChEMBL values, standardized by MDeePred [26], were used as indicators of the bioactivity strength of compounds against their targets [35].

As illustrated in Fig. 7, most existing models exhibited limited capability to infer binding strength, as they were trained solely on binarized interaction labels. In contrast, EnsDTI's predictions showed a weak yet positive correlation with pChEMBL values, outperforming other models. This result highlights EnsDTI's potential to approximate true bioactivity data, making it a promising tool for practical drug development applications.

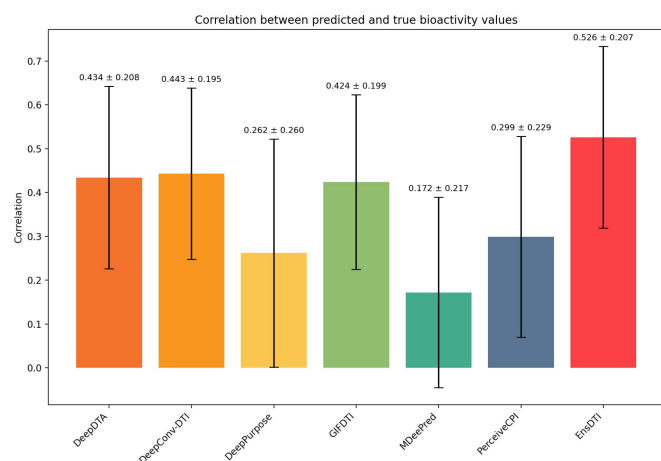


Fig. 7. Pearson correlation between predicted probabilities and bioactivity pChEMBL values. EnsDTI demonstrates a higher correlation with experimental bioactivity data compared to baseline models, showcasing its potential to approximate true binding affinities despite being trained on binarized interaction labels. Details in Section V-D.

We further investigated whether our model could robustly approximate true binding affinities for two functionally similar proteins. Using Ephrin type-B receptor 1 (CHEMBL5072) and Ephrin type-A receptor 4 (CHEMBL3988) as examples, both proteins are receptor tyrosine kinases that interact with membrane-bound ephrin ligands on adjacent cells. The kinome database includes interaction records for CHEMBL5072 and CHEMBL3988 with 33 and 28 small molecules, respectively, among which 22 compounds are shared by both proteins. To assess the predictive performance, we generated regression scatter plots comparing the predicted probabilities and experimental affinity values for each model on CHEMBL5072 and CHEMBL3988. EnsDTI was compared against GIFDTI and DeepPurpose, which performed most closely to EnsDTI as indicated in Table II.

As shown in Fig. 8, EnsDTI's predictions for candidate drugs targeting CHEMBL5072 and CHEMBL3988 demonstrated the highest correlation with experimental assay data. In contrast, other models displayed weaker correlations with bioactivity pChEMBL values and lacked robustness when the target protein changed. Notably, base classifiers frequently misclassified small molecules with high bioactivity—those more likely to be potential drugs—as negative samples, underscoring limitations in practical drug discovery. EnsDTI, with significantly fewer misclassifications, consistently achieved higher correlation with actual bioactivity values, highlighting its capacity to approximate true binding affinities and its potential applicability in drug development.

E. Ablation Study

In constructing EnsDTI, the number and selection of expert models are crucial considerations. Is it necessary to include exactly six experts in the mixture-of-experts (MoE) framework? Which combinations can achieve comparable performance with

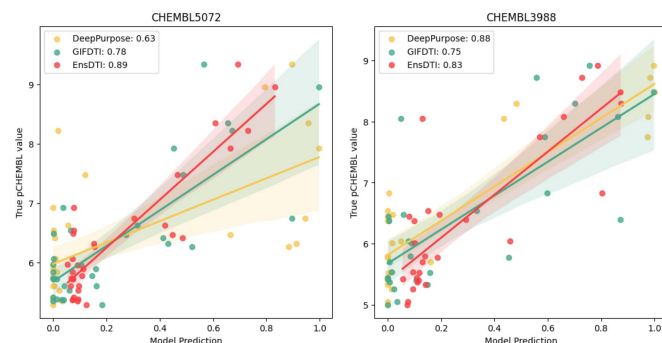


Fig. 8. Regression scatter plots for candidate drugs targeting CHEMBL5072 and CHEMBL3988. The plots compare the predicted probabilities and experimental pChEMBL values generated by EnsDTI (red), GIFDTI (green), and DeepPurpose (yellow). Correlation coefficients between model predictions and experimental pChEMBL values are provided in the legend. The results show that EnsDTI achieves the highest correlation with experimental bioactivity values, highlighting its robustness and accuracy. Notably, EnsDTI demonstrates fewer misclassifications, particularly for compounds with high bioactivity, even across proteins with similar functional profiles.

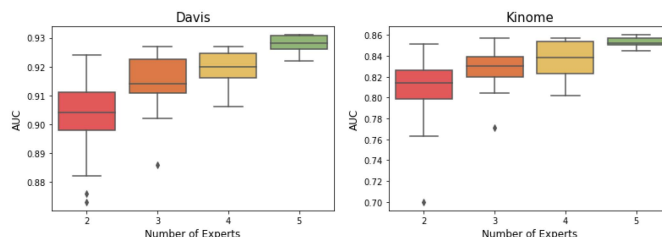


Fig. 9. Performance of EnsDTI variants. We conducted ablation study to analysis how number of experts effect the performance of EnsDTI, details in Section V-E.

reduced training costs? To address these questions, we systematically ablated EnsDTI by incrementally removing individual DTI expert models and trained all possible ablation versions under the same experimental conditions. The complete performance results are provided in Supplementary Materials. Based on these experiments, we plotted boxplots (Fig. 9) to visualize the impact of the number of expert models on EnsDTI's performance. The results show a clear trend: as the number of expert models increases, EnsDTI achieves higher performance with reduced variance and bias.

Interestingly, the impact of the number of experts varies across datasets. For instance, in the Davis dataset, combining two experts yields a slight improvement in predictive performance, though the gain is limited. In contrast, on the Human dataset, reducing the MoE framework to only two experts can negatively affect performance; for example, an EnsDTI variant constructed using DeepDTA and GIFDTI performs less favorably than either model alone. These observations indicate that the effectiveness of the MoE framework may depend on the degree of agreement among individual models, where substantial disagreement can reduce ensemble consistency. Notably, when the number of experts reaches four or more, most ablated versions of EnsDTI outperform the individual models in terms of predictive accuracy.

TABLE IV

PERFORMANCE COMPARISON OF ENSDTI AND DLM-DTI ACROSS FOUR BENCHMARK DATASETS. ENSDTI ACHIEVES COMPARABLE PERFORMANCE TO THE PRETRAINED DLM-DTI MODEL, WHICH LEVERAGES CHEMBERTA AND PROTBERT PRETRAINED ON EXTENSIVE MOLECULAR AND PROTEIN DATASETS. DESPITE LACKING SUCH PRETRAINING, ENSDTI DEMONSTRATES ITS EFFECTIVENESS AND APPLICABILITY ACROSS DIVERSE DTI BENCHMARKS.

Model	davis			kiba			kinome			human		
	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC	ACC	AUROC	AUPRC
DLM-DTI	0.939	0.939	0.687	0.878	0.910	0.9732	0.751	0.813	0.859	0.937	0.983	0.985
EnsDTI	0.949	0.929	0.670	0.906	0.942	0.958	0.795	0.866	0.790	0.959	0.958	0.977

F. Compared With Pre-Trained Model

To further validate the effectiveness and applicability of EnsDTI, we compared it with the state-of-the-art pretrained model, DLM-DTI [36]. DLM-DTI employs pretrained ChemBERTa [3] as the drug encoder and uses pretrained ProtBERT [37] for distilling target embeddings. ChemBERTa is pretrained on 77 M molecular SMILES from PubChem, while ProtBERT is pretrained on up to 393 billion amino acids. We conducted experiments on the Davis, KIBA, Kinome, and Human databases using the code and parameters provided by the authors of DLM-DTI. The dataset partitioning was consistent with that used in all other experiments. The experimental results are presented in Table IV. Our experiments demonstrate that EnsDTI achieves comparable performance to DLM-DTI, even without extensive pretraining on large datasets.

VI. DISCUSSION

Our experimental results across four benchmark datasets demonstrate that EnsDTI achieves consistent and measurable gains over existing DTI models. The Davis, KIBA, Kinome, and Human datasets differ markedly in label balance, bioactivity distributions, compound diversity, and target family composition; consequently, single-model architectures often overfit to dataset-specific patterns and exhibit unstable behavior. By contrast, EnsDTI aggregates experts with heterogeneous inductive biases, ranging from sequence-based encoders to graph and convolutional models, allowing the gating network to exploit complementary strengths while mitigating individual weaknesses. This design results in more stable predictions across diverse biochemical contexts. For instance, on the Davis dataset, EnsDTI improves AUPRC from 0.636 to 0.670, and on KIBA it increases AUROC from 0.915 to 0.942. Importantly, this robustness extends to challenging cold-start scenarios: when 20% of drugs were held out as unseen, EnsDTI maintains higher AUROC and AUPRC than all baseline models, indicating its ability to generalize to novel compounds while preserving reliable predictions.

These improvements highlight two key advantages of EnsDTI. First, the mixture-of-experts architecture effectively aggregates complementary strengths across models trained with diverse feature modalities and encoding strategies. This mechanism mitigates the dataset-specific weaknesses observed in individual experts and reduces performance variance across heterogeneous benchmarks. Second, the inductive conformal prediction (ICP) module provides calibrated confidence estimates, enabling EnsDTI to couple prediction accuracy with reliability.

Our confidence-stratified analysis shows that EnsDTI's error rate decreases monotonically as prediction confidence increases, achieving error rates below 1% at confidence thresholds above 0.6 for three of the four datasets.

Furthermore, EnsDTI demonstrates an ability to approximate underlying bioactivity signals despite being trained on binarized labels. The model achieves the highest Pearson correlation with pChEMBL values among all baselines, with a relative improvement of 14.5% over the SOTA model. This trend is reinforced in case studies involving functionally similar kinase proteins, where EnsDTI exhibits strong correlation with experimental affinities and reduces false negative predictions for highly active compounds.

Overall, these findings indicate that EnsDTI not only improves predictive performance in absolute and relative terms, but also enhances robustness, confidence calibration, and biological relevance. These properties collectively suggest that EnsDTI can serve as a reliable and practical tool for ranking and prioritizing candidate molecules in early-stage drug discovery pipelines.

VII. CONCLUSION

In this work, we introduce EnsDTI, a novel deep learning framework for drug-target interaction (DTI) prediction, leveraging a mixture-of-experts (MoE) architecture to address the limitations of existing DTI models. While current deep learning approaches often excel in specific data distributions, they struggle to deliver consistent predictions across unseen data points due to independently learned latent spaces. EnsDTI overcomes these challenges by integrating diverse features learned from multiple DTI expert models, providing reliable and accurate rankings of candidate drug-target pairs.

The architecture of EnsDTI enhances predictive performance through a modular process: first, each DTI expert model is trained to its maximum potential. Next, the experts are frozen to generate prediction probabilities, which are subsequently processed by a gating network to combine their knowledge. Finally, an inductive conformal predictor (ICP) module assigns confidence scores to predictions, offering a quantifiable measure of reliability without altering the core predictions. This layered design ensures robust predictions, even for unseen drug-protein pairs.

Evaluations on four widely used DTI benchmark datasets demonstrate that EnsDTI achieves improved predictive performance compared to existing methods. In addition to higher accuracy, the model generates calibrated confidence estimates

for individual predictions, supporting more informed decision-making. Notably, the interaction rankings produced by EnsDTI exhibit strong correlation with experimentally derived docking affinities, indicating its potential utility in structure-based drug discovery. By jointly providing prediction scores and uncertainty estimates, EnsDTI offers a reliable approach for prioritizing candidate compounds in large-scale screening settings, with practical relevance to early-stage drug discovery applications.

ACKNOWLEDGMENT

The ICT at Seoul National University provides research facilities for this study. The authors used ChatGPT for assistance with language editing and expression refinement during the manuscript revision. All technical content, analyses, and conclusions are the original work of the authors and have been independently verified for accuracy. Additionally, we extend our gratitude to Sungjoon Park, Jungseob Yi, Chang-Yun Cho, and Sangsoo Lim for their valuable discussions.

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Yijiangxiu Lu received the BSEE degree from the University of Science and Technology Beijing, China. She is currently working toward the PhD degree in computer science with Seoul National University, South Korea. She is with Bio and Health Informatics Lab under the supervision of professor Sun Kim. Her research interests include developing machine learning tools to predict complex heterogeneous relations among multi-type molecules, such as proteins and drugs, under diverse conditions and environments.



Soosung Kang received the PhD degree in organic and medicinal chemistry from Purdue University. He is currently a professor with the College of Pharmacy, Ewha Womans University, South Korea. He conducted postdoctoral research on drug development for neurodegenerative diseases with Northwestern University. His research interests include developing therapeutics targeting proteins associated with oxidative stress in mitochondria and neurodegenerative diseases, utilizing medicinal, organic, computational chemistry, and artificial intelligence.



Sangseon Lee received the PhD degree in computer science and engineering from Seoul National University, in 2020. He is currently an Assistant Professor with the Department of Artificial Intelligence, Inha University, South Korea. His research interests include developing deep learning and machine learning models for analyzing graph-structured and multi-omics data. His work aims to interpret the biological mechanisms of living organisms and model molecular graphs by incorporating subgraph structures and 3D information.



Sun Kim received the PhD degree in computer science from the University of Iowa, in 1997. He held faculty positions with Indiana University and was a senior computer scientist with DuPont Central Research. He is currently a full professor with the Department of Computer Science and Engineering, Seoul National University, South Korea, where he also directs the Bioinformatics Lab. He is also the CTO with AIGENDRUG Company, Ltd. His research interests include bioinformatics, health informatics, and the development of computational methods for

biomedical applications, focusing on advanced machine learning and data-driven analysis.